

The p110 δ structure: mechanisms for selectivity and potency of new PI(3)K inhibitors

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Deregulation of the phosphoinositide-3-OH kinase (PI(3)K) pathway has been implicated in numerous pathologies including cancer, diabetes, thrombosis, rheumatoid arthritis and asthma. Recently, small-molecule and ATP-competitive PI(3)K inhibitors with a wide range of selectivities have entered clinical development. In order to understand the mechanisms underlying the isoform selectivity of these inhibitors, we developed a new expression strategy that enabled us to determine to our knowledge the first crystal structure of the catalytic subunit of the class IA PI(3)K p110 δ . Structures of this enzyme in complex with a broad panel of isoform- and pan-selective class I PI(3)K inhibitors reveal that selectivity toward p110 δ can be achieved by exploiting its conformational flexibility and the sequence diversity of active site residues that do not contact ATP. We have used these observations to rationalize and synthesize highly selective inhibitors for p110 δ with greatly improved potencies.

The PI(3)Ks are structurally closely related lipid kinases that catalyze the ATP-dependent phosphorylation of phosphoinositide substrates^{1,2}. Together with the serine/threonine protein kinase B (PKB), PI(3)Ks constitute a central signaling hub that mediates many diverse and crucial cell functions, including cell growth, proliferation, metabolism and survival^{1,3}. The observation that PI(3)Ks acting downstream of receptor tyrosine kinases (RTKs) are the most commonly mutated kinases in human cancers has spurred interest in understanding the structural mechanisms for how these mutations upregulate PI(3)K activity and in developing selective and drug-like PI(3)K inhibitors^{4,5}.

PI(3)Ks can be grouped into three classes based on their domain organization⁶. Class I PI(3)Ks are heterodimers consisting of a p110 catalytic subunit and a regulatory subunit of either the 'p85' type (associated with the class IA p110 isoforms p110 α , p110 β or p110 δ) or the 'p101' type (associated with the class IB p110 isoform p110 γ). The p110 catalytic subunit consists of an adaptor-binding domain (ABD), a Ras-binding domain (RBD), a C2 domain, a helical domain and the kinase domain^{7–10}.

Mutant mice and inhibitor studies have shown less functional redundancy for the various class I PI(3)K isoforms than previously anticipated. Whereas p110 α and p110 β are ubiquitously expressed, p110 γ and p110 δ are predominantly found in hematopoietic cells^{11–13}. Genetic deregulation of PI(3)K activity (oncogenic gain-of-function mutations, overexpression) has been implicated in cancer (all class I PI(3)K isoforms)^{14–17}, diabetes (p110 α)¹⁸, thrombosis (p110 β)¹⁹, rheumatoid arthritis (p110 γ and p110 δ)²⁰ and asthma (p110 γ and p110 δ)^{21,22}. Consequently, the selective inhibition of individual PI(3)K isoforms using small-molecule and ATP-competitive inhibitors is a promising therapeutic strategy²³. However, because all active site side chains in contact with ATP are completely conserved throughout all class I PI(3)K family members (Supplementary Fig. 1), this remains a challenging objective. Furthermore, in order to minimize undesired and often poorly understood toxic side effects, such inhibitors ideally would have to show no cross-reactivity toward off-pathway targets²⁴.

The earliest generation of small-molecule and ATP-competitive PI(3)K inhibitors, including the pan-selective LY294004 (ref. 25) and wortmannin²⁶, were important tools for investigating PI(3)K-mediated cellular responses in the laboratory, but their low affinity (LY294002), instability (wortmannin), nonselectivity and toxicity limited their clinical use. However, further chemical modifications of some of these early inhibitors significantly helped to improve their drug-like properties. For example, PWT-458 (Wyeth) and PX-866 (Oncothyreon) are modified wortmannin-based PI(3)K inhibitors with improved pharmacological properties that are currently in phase 1 clinical trials^{27,28}.

The first crystal structures of p110 γ in complexes with pan-selective PI(3)K inhibitors²⁹ made it possible to begin to rationalize PI(3)K isoform-selective inhibitors such as AS604850 (Merck-Serono) for p110 γ (ref. 30). However, many of these inhibitors retained off-target activities, and, partially due to the lack of crystal structures of other PI(3)K isoforms and PI(3)K-related protein kinases (PIKKs), these unwanted side effects were difficult to rationalize.

Notably, the development of multi- and pan-selective PI(3)K inhibitors as well as dual PI(3)K/mTOR or PI(3)K/tyrosine kinase inhibitors³¹ rather than isoform-selective PI(3)K inhibitors remains a valid therapeutic strategy. XL-147 (Exelixis), which is currently evaluated in combination with other cancer therapeutics, is in phase 1 and 2 clinical trials for the treatment of non-small-cell lung cancer. GDC-0941 (Roche)³² is also in phase 1 trials for the treatment of breast cancer³³; both XL-147 and GDC-0941 are examples of pan class I-selective PI(3)K inhibitors. NVP-BEZ235 (Novartis), which is currently in phase 1 and 2 trials for breast cancer³⁴, and SF1126 (Semaphore), an RGDS peptide-conjugated prodrug of LY294002 presently in phase 1 trials³⁵, are examples of dual-selectivity PI(3)K/mTOR inhibitors.

Recently, several new class I PI(3)K isoform-selective inhibitors showing improved selectivities and potencies have been reported, and some of them have entered clinical trials: CAL-101 (Calistoga), a derivative of the highly p110 δ -selective inhibitor IC87114 (ref. 36) with increased potency, entered phase 1 for the treatment of acute

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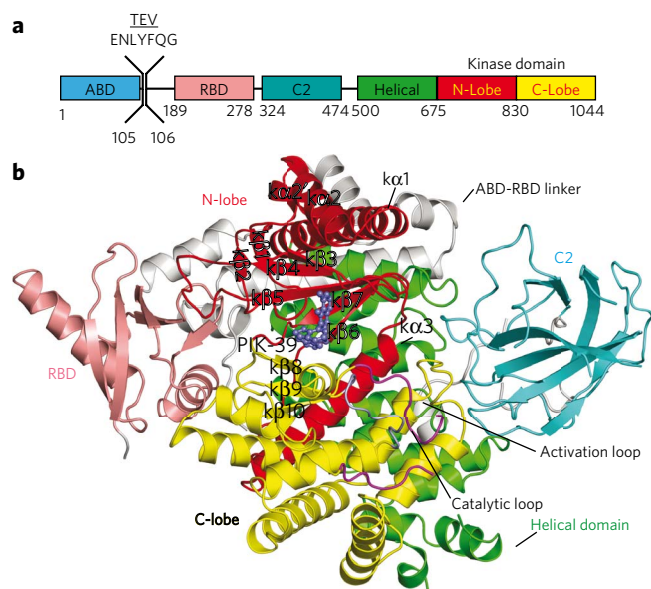


Figure 1 | Domain organization, construct design and overall crystal structure of p110 δ . (a) A TEV protease cleavage site was introduced between residues 105 and 106 of the p110 δ ABD-RBD linker. The numbers below the boxes correspond to the indicated domain boundaries. After purification of the p110 δ -iSH2 complex, the catalytic core is released by cleavage with TEV protease. (b) Cartoon representation of the overall cocrystal structure of the Δ ABDp110 δ -PIK-39 complex. Linker regions are colored in white, the RBD in salmon, the C2 domain in cyan, the helical domain in green, the kinase domain N-lobe in red and the kinase domain C-lobe in yellow. PIK-39 is shown in light blue as a ball-and-stick representation. Selected secondary structure elements of the kinase domain are labeled.

myeloid leukemia (AML) and B-cell chronic lymphoid leukemia (CLL). The p110 β -selective AZD6482 (AstraZeneca) is in phase 1 trials for the treatment of thrombosis. Notably, however, despite a growing list of such isoform-selective compounds, little is known about what determines isoform selectivity on a structural level.

Impaired PI(3)K δ signaling results in severe defects of innate and adaptive immune responses, suggesting that targeting of this isoform would be a beneficial therapeutic strategy^{20,24}. To elucidate the molecular mechanisms of isoform selectivity of PI(3)K δ inhibitors, we report the crystal structure of the p110 δ catalytic core, both free and in complexes with a broad panel of new and mostly p110 δ -selective PI(3)K inhibitors. Our study provides the first detailed structural insights into the active site of a class IA PI(3)K occupied by noncovalently bound inhibitors. Furthermore, our structures suggest mechanisms to achieve p110 δ selectivity and to increase the potency of inhibitors without sacrificing isoform selectivity. To obtain these structures, we developed a unique expression and purification scheme that has now been extended to all class IA PI(3)K isoforms.

With our new set of p110 δ crystal structures and better models of flexibility resulting from molecular dynamics simulations, we are now starting to understand why p110 δ can be more easily deformed to open an allosteric pocket in which p110 δ -selective inhibitors can be accommodated.

RESULTS

Expression and catalytic activity of Δ ABDp110 δ

Our initial attempts to express either the full-length or the ABD-truncated p110 δ catalytic subunit in Sf9 cells produced only insoluble protein. However, we could readily express and purify p110 δ in complexes with only the iSH2 domain of p85 α . We devised a new expression and purification strategy by introducing a TEV protease

cleavage site in the linker region between the ABD and the RBD of p110 δ (Fig. 1a) with the objective of generating an ABD-truncated version of this isoform for crystallization trials. The Δ ABDp110 δ construct showed a considerably enhanced lipid kinase activity *in vitro* when compared with either the holo p110 δ -p85 α complex or the p110 δ -p85 α nicSH2 complex (Supplementary Fig. 2).

Overall structure of Δ ABDp110 δ

Crystallographic statistics for all p110 δ datasets are given in Supplementary Table 1. The overall fold of p110 δ is very similar to the catalytic subunits of p110 γ and p110 α (Fig. 1b)^{8,37}. Helices L α 1–L α 3 of the ABD-RBD linker pack tightly against the helical domain, and stretch from the RBD to the C2 domain. The helices $\kappa\alpha$ 1 and $\kappa\alpha$ 2/ $\kappa\alpha$ 2' of the kinase domain form a hairpin in the N-lobe that sits on top of a five-stranded β -sheet formed by $\kappa\beta$ 3– $\kappa\beta$ 7, and this hairpin structurally distinguishes PI(3)Ks from protein kinases. These latter helices seem to extend the antiparallel A/B pairs of α -helices found in the helical domain. The kinase domain has an extensive, tightly packed interface with the helical domain. All of the catalytically important motifs within this domain are well ordered, with the exception of residues 920–928 of a region known as the “activation” or phosphoinositide-binding loop. Notably, the residues within the p110 δ 893-DRH-895 motif located in the “catalytic” loop, a motif conserved in all PI(3)Ks and inverted (HRD) in protein kinases, adopt a different conformation from what was previously observed in the structure of p110 γ (Supplementary Fig. 3)⁸. This different conformation might be critical for the correct positioning of the DFG aspartate at the beginning of the activation loop.

All the domains of p110 δ superimpose closely on previously reported PI(3)K structures (Supplementary Fig. 4a–f). However, the most noteworthy difference in the overall structure of p110 δ relative to p110 α or p110 γ is a change in the orientation of the N-lobe with respect to the C-lobe of the kinase domain. This shift may reflect motions characteristic of the catalytic cycle, analogous to the hinging and sliding motions of the N- and C-lobes that have been described for protein kinases³⁸. Furthermore, the RBD shifts relative to the N-lobe of the kinase domain (Supplementary Fig. 4g). The RBD mediates interaction with Ras in a GTP-dependent manner for all three isoforms^{11,12,39,40}. Despite the great sequence divergence among the isoforms in the RBD, the overall RBD backbone conformation is very closely preserved among the various class I isoforms (Supplementary Fig. 4f). However, differences in the orientation of the RBD relative to the kinase domain suggest the possibility of different mechanisms of activation by Ras. The conformation of the loop connecting $\kappa\beta$ 4 and $\kappa\beta$ 5 (Tyr763 to Val774 in p110 δ) in the N-lobe is different in all the isoforms (longest in p110 α , shortest in p110 δ), and this correlates with the orientation of the RBD. Within the RBD of p110 δ , residues 231–234 are disordered. The equivalent region in p110 α is an ordered helix (R α 2), whereas in p110 γ this region is ordered only in the Ras-p110 γ complex, although it has a completely different conformation than in p110 α .

Cocrystallization of p110 δ with inhibitors

We chose a set of chemically diverse inhibitors in order to understand the structural mechanisms that underlie p110 δ -specific inhibition in contrast to broadly specific PI(3)K inhibitors. Even though we obtained crystals grown in the presence of ATP, only a weak density somewhat larger than what would be expected for an ordered water molecule was observed in the hinge region. We will refer to this structure as the apo form of p110 δ .

ATP-binding pocket

All of the compounds presented here contact a core set of six residues in the ATP-binding pocket (Supplementary Table 2), and—apart from the hinge residue Val827 in p110 δ —these residues are invariant in all of the class I PI(3)K isoforms. Based on our

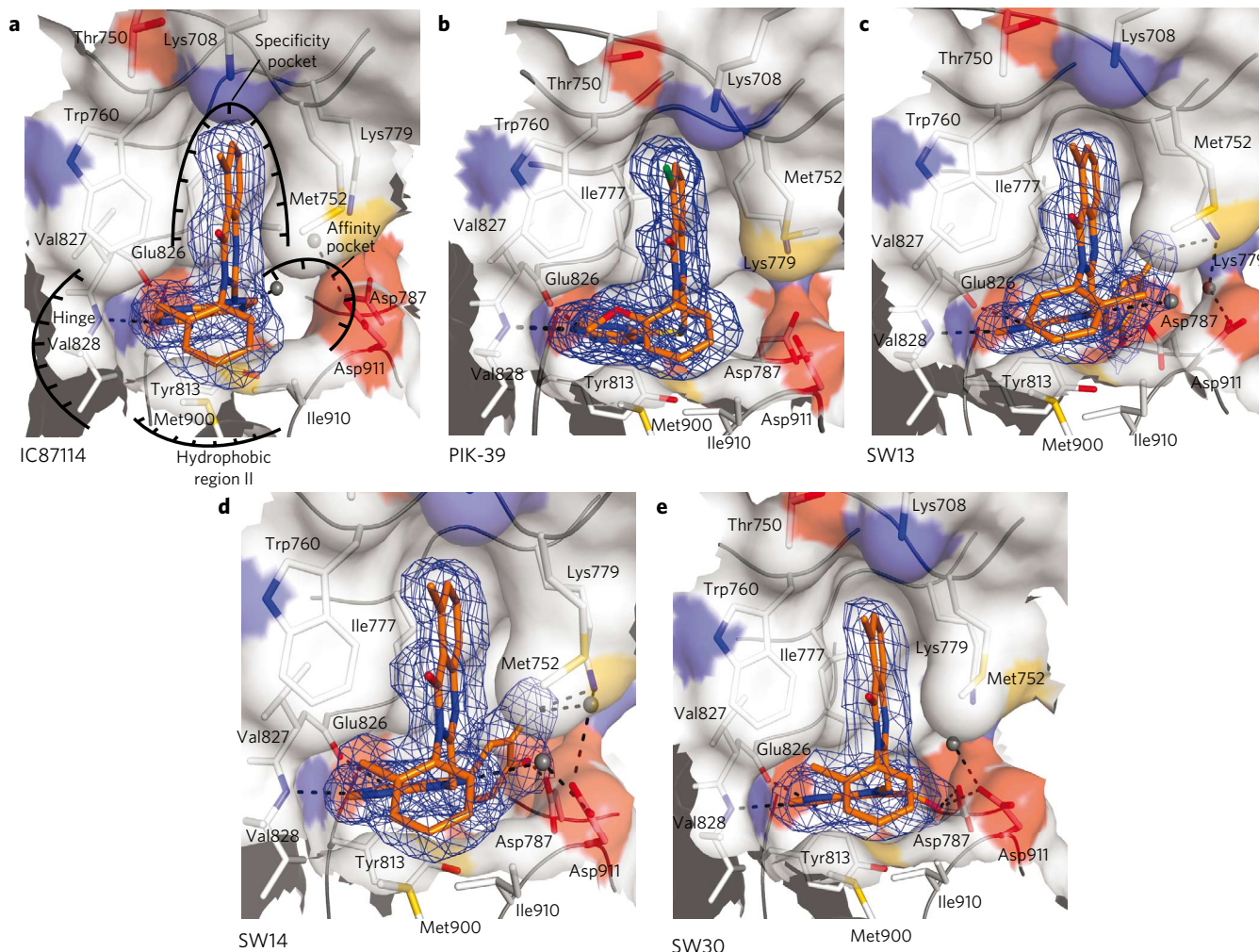


Figure 2 | The propeller-shaped p110 δ -selective inhibitors induce the formation of the specificity pocket. (a–e) Shown are the active sites of p110 δ in complex with the inhibitors IC87114 (**a**), PIK-39 (**b**), SW13 (**c**), SW14 (**d**) and SW30 (**e**). Key residues that outline the active site and interact with the compounds and the $2mF_o - DF_o$ electron densities (contouring level 1σ) are presented. Selected water molecules in the active sites are shown as gray spheres. Note that IC87114 and PIK-39 do not fill the affinity pocket, whereas SW13, SW14 and SW30 do. Dashed black lines represent hydrogen bonds.

inhibitor-bound structures of p110 δ as well as previously described PI(3)K complexes^{18,29,30,32,41}, we can define four regions within the ATP-binding pocket that are important for inhibitor binding (Fig. 2a): an “adenine” pocket (hinge), a “specificity” pocket, an “affinity” pocket and the hydrophobic region II located at the mouth of the active site^{18,42}. Of the core active site residues, only two are in contact with inhibitors in all complexes: Val828 and Ile910. Residues 825–828 line the adenine pocket and form a hinge between the N-lobe and C-lobe of the catalytic domain. The backbone amide of the hinge Val828 makes a characteristic hydrogen bond in all of the p110-inhibitor complexes. Additionally, the backbone carbonyl of hinge Glu826 establishes hydrogen bonds to most of the inhibitors.

Our selection of inhibitors can be organized into three types. The first type includes inhibitors that adopt a propeller-shaped conformation (two roughly orthogonally oriented aromatic ring systems) when bound to the enzyme (Fig. 2a–e and Supplementary Fig. 5). These are mostly p110 δ -selective inhibitors that stabilize a conformational change that opens a hydrophobic “specificity” pocket in the active site that is not present in the apo structure of the enzyme as previously reported for the p110 γ -PIK-39 crystal structure¹⁸. We also cocrystallized the p110 δ enzyme with a second set of mostly flat and multi- to pan-selective class I PI(3)K inhibitors that do not provoke such a conformational rearrangement. AS15, which has a distorted propeller shape when bound to

the enzyme, is the only example of a third type of inhibitor that is highly selective for the p110 δ isoform, although it does not open the specificity pocket.

The propeller-shaped inhibitors IC87114 and PIK-39

The discovery of the p110 δ -selective inhibitor IC87114 (ICOS) in 2003 (ref. 36) was a proof of principle that isoform selectivity of PI(3)K inhibitors can be accomplished, and so far it remains one of the most selective p110 δ inhibitors known.

The crystal structures of the p110 δ -IC87114 (**1**) (Fig. 2a) and the p110 δ -PIK-39 (**2**) (Fig. 2b) complexes show that the purine group of the compounds resides within the adenine pocket and establishes hydrogen bonds to the hinge residues Glu826 and Val828. The quinazolinone moiety is sandwiched into the induced hydrophobic specificity pocket between Trp760 and Ile777 on one side and two P-loop residues, Met752 and Pro758, on the other side. The specificity pocket is not present in the apo enzyme where the P-loop Met752 rests in its “in” position leaning against Trp760. The toluene group (IC87114) and the methoxyphenyl group (PIK-39) attached to the quinazolinone moiety project out of the ATP-binding pocket over a region that we will refer to as hydrophobic region II.

PIK-39 binding to both p110 δ and p110 γ induces a slight opening in the ATP-binding pocket. The p110 δ ATP-binding pocket accommodates the PIK-39-induced conformational change by a

local change in the conformation of the P-loop (residues 752–758 in p110 δ), whereas the equivalent opening of the p110 γ pocket is accompanied by a conformational change that involves much of the N-lobe moving with respect to the C-lobe. The loop between $\kappa\alpha 1$ and $\kappa\alpha 2$ of p110 γ (residues 752–760) sits on top of the P-loop (residues 803–811) and appears to rigidify it, so that the compound-induced opening of the pocket is accompanied by a shift of the N-lobe as a unit (**Supplementary Fig. 6** and **Supplementary Movies 1** and **2**). In contrast to p110 γ , in p110 δ the slightly shorter $\kappa\alpha 1$ – $\kappa\alpha 2$ loop leaves the P-loop largely free and able to move independently of the rest of the N-lobe. We proposed that opening of the specificity pocket might be easier in p110 δ compared to p110 γ .

Molecular dynamics simulations of p110 δ and p110 γ

Perturbation analysis by molecular dynamics simulations suggests that the free energy of the specificity pocket closure is more favorable in p110 γ than p110 δ (**Supplementary Fig. 7**). To quantify the higher degree of flexibility within the p110 δ active site, we performed molecular dynamics simulations of the apo enzymes of both isoforms (see **Supplementary Methods** and **Supplementary Movies 3** and **4**). The potential energy of the interaction of PIK-39 with the enzyme is more favorable for p110 δ than for p110 γ

(**Supplementary Fig. 8**). Our results further show that the distance between Trp760 (Trp812 in p110 γ) and the P-loop Met752 (Met804 in p110 γ) does not change appreciably in p110 δ over the course of the simulation because the conformational changes observed for both residues are synchronized with each other—that is, the tryptophan smoothly follows the methionine and vice versa. In contrast, in p110 γ , as the Met804 transiently assumes alternate rotamers, it briefly creates gaps between itself and Trp812. Trp812 of p110 γ is sterically constrained by a hydrogen bond to Glu814 (Met762 in p110 δ) and is therefore unable to flex in synchrony with Met804 as in p110 δ . Additionally, in p110 γ there is a more pronounced hydrophobic interaction between the Trp812 and the hinge Ile881, which might further restrain the position of the tryptophan. The transient opening of the specificity pocket in p110 γ would allow water to become trapped, leading to an unfavorable entropy change.

Increased potencies of propeller-shaped inhibitors

The SW series (**3**, **4** and **5**)⁴³ and INK series (**6** and **7**) of inhibitors take advantage of both the specificity pocket and the affinity pocket (synthesis details for these compounds are given in the **Supplementary Methods**). This pocket is lined by a thin hydrophobic strip formed by Leu784, Cys815 and Ile825 at the back of

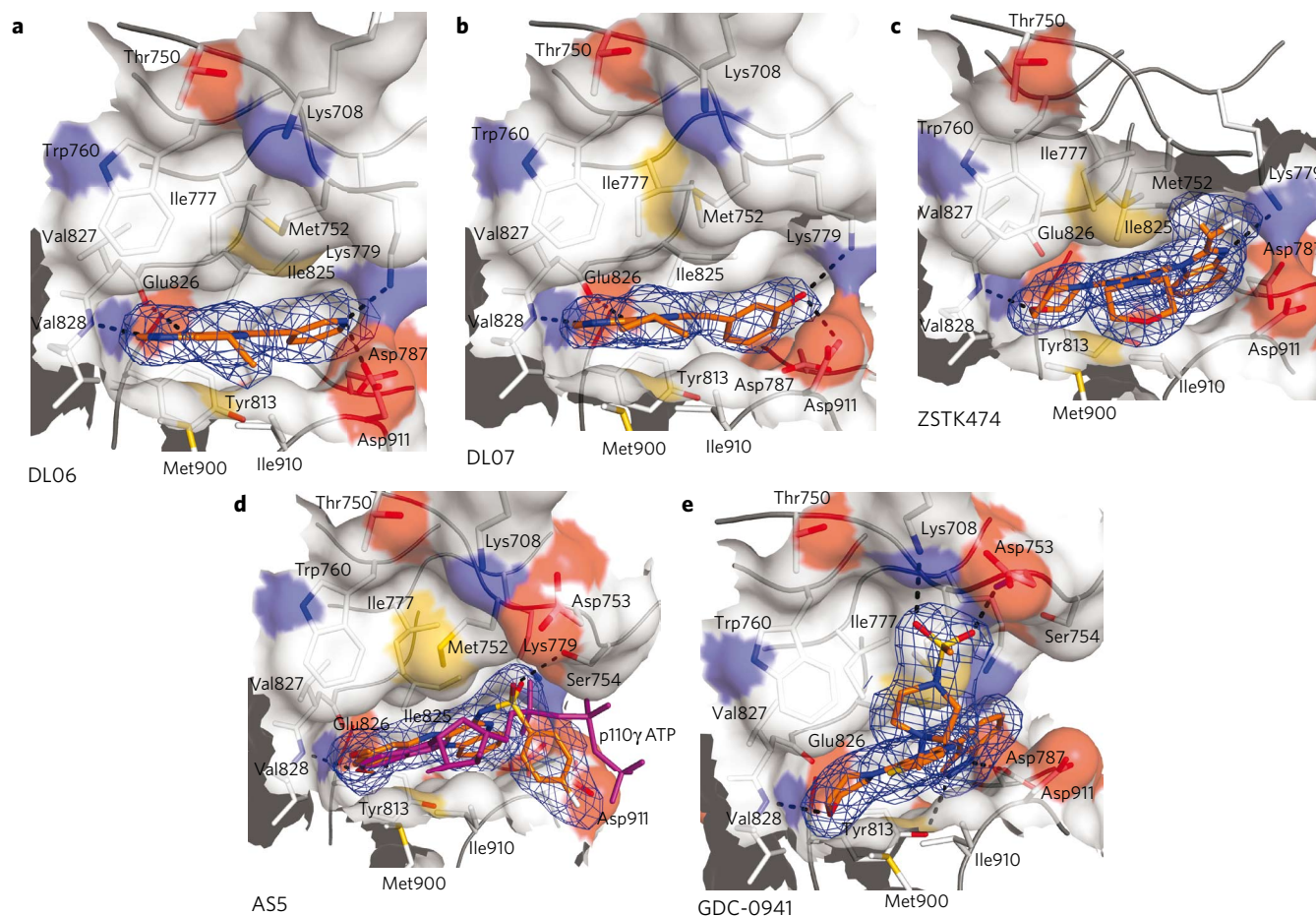


Figure 3 | The flat inhibitors DL06, DL07, ZSTK474, AS5 and GDC-0941 are multi- to pan-selective class I PI(3)K inhibitors that do not induce the opening of the specificity pocket. (a–e) Shown are the binding modes of DL06 (a), DL07 (b), ZSTK474 (c), AS5 (d) and GDC-0941 (e) in the active site of p110 δ . Met752 is in its ‘in’ position for all these compounds. For d, the structure of the p110 γ -ATP complex (Protein Data Bank entry 1E8X) was superimposed on the C α -backbone of p110 δ to show the proximity of the sulfonyl group of AS5 to the α -phosphate group of ATP (purple). This sulfonyl group is a hydrogen bond acceptor to Ser754 located in the P-loop of p110 δ . (e) GDC-0941 is a pan-class IA PI(3)K inhibitor that (like AS15) interacts with residues outside the active site. GDC-0941 occupies the adenine pocket and the affinity pocket within the active site of p110 δ and engages there in hydrogen bonds with Val828, Tyr813 and Asp787. Additionally, the substituted piperazine group of GDC-0941 extends out of the ATP-binding site where its methylsulfonyl moiety acts as a hydrogen bond acceptor for Asp753 of the P-loop and Lys708 at the beginning of $\kappa\alpha 2$. The contouring level of the $2mF_o - DF_c$ electron densities is 1σ for each compound.

the ATP-binding pocket and flanked on the top by the side chains of Pro758 and Lys779 and on the bottom by Asp787 (hydrophobic region I in protein kinases). These mostly p110 δ -selective compounds (SW14 is dual-selective for p110 γ and p110 δ) are also propeller-shaped, but they have additional decorations when compared to IC87114 and PIK-39 in the form of an *ortho*-fluorophenol (SW14), a *para*-fluorophenol (SW13) or a butynol group (SW30) attached to the central pyrazolopyrimidineineamine scaffold (Fig. 2c–e). These groups explore the affinity pocket where they engage in hydrogen bonds with Asp787 (SW13, SW14 and SW30) and Lys779 (SW13 and SW14). Additionally, the butynol OH group of SW30 also serves as a hydrogen bond donor to the DFG Asp911 at the start of the activation loop, and the phenolic OH group of SW13 engages in hydrogen bonding with Tyr813. This set of new inhibitor-enzyme interactions leads to a substantial increase in the inhibitors' potencies toward p110 δ , which is reflected in their greatly lowered half-maximal inhibitory concentration (IC₅₀) values (Supplementary Tables 2 and 3). The propeller shape of a compound alone does not guarantee p110 δ specificity, as shown by INK666 (Supplementary Fig. 5b).

Our structures of p110 δ in complex with SW13, SW14 and SW30 also indicate a conformational flexibility for the catalytic DFG Asp911. This residue assumes two alternative conformations in the p110 δ -SW structures. One of these, the 'in' conformation, coincides with its putative ATP/Mg²⁺-binding position (based on the p110 γ -ATP complex). The other conformation has the DFG Asp911 swung away ('out' conformation). In the p110 δ -SW14 and p110 δ -SW30 structures, DFG Asp911 is found in the 'out' conformation, whereas in the p110 δ -SW13 complex it is 'in'. In protein kinases, a shift of the DFG aspartate from the 'in' conformation (ATP-bound) to the 'out' conformation is characteristic of the catalytic cycle. By analogy, it may be that these inhibitors are inducing conformations characteristic of the PI(3)K catalytic cycle.

p110 δ complexes with flat and multiselective inhibitors

ZSTK474 (ref. 44) (8), DL06 (9), DL07 (10), AS5 (11) and GDC-0941 (ref. 32) (12) are fairly flat compounds that do not open the specificity pocket and achieve relatively little isotype selectivity. Their binding provokes some motions of the P-loop side chains of p110 δ , and these conformational changes are coordinated with changes in the conformation of the DFG Asp911 in the C-lobe.

A minimalistic approach to achieve PI(3)K inhibition

The DL06/07 series of PI(3)K inhibitors can best be described as pan-selective p110 inhibitors that represent a minimalistic approach

to achieve PI(3)K inhibition (see **Supplementary Methods** for synthesis details). They are flat and small compounds with a minimal design that is just sufficient to span the adenine pocket via their pyrazolopyrimidine moiety and project into the affinity pocket by means of a phenol (DL07) or a pyridine (DL06) group attached to a propyne "stick" (Fig. 3a,b). The DL07 phenol group interacts with the DFG Asp911, forcing it to its 'in' conformation. It also induces rotations in the side chain of P-loop Met752, but not to its 'out' conformation. Similar interactions are formed by DL06.

p110 δ -ZSTK474

The new pan-selective triazine PI(3)K inhibitor ZSTK474 strongly inhibits the growth of tumor cells in human cancer xenografts and therefore is a potential candidate for further clinical development⁴⁴. Its crystal structure in complex with p110 δ shows it flipped over relative to what was predicted in a computational p110 γ -ZSTK474 model⁴⁴ (Fig. 3c). The oxygen of one of the morpholino groups is positioned as the hinge hydrogen bond acceptor, and the morpholino ring adopts a chair conformation. The benzimidazole group extends into the affinity pocket where its nitrogen acts as a hydrogen bond acceptor for the primary amine of Lys779. The difluoromethyl group points toward Pro758 in the upper wall of the hydrophobic affinity pocket. The second morpholino group adopts a somewhat twisted chair conformation and projects out of the ATP-binding pocket in the same manner as the phenyl group of LY294002 where it occupies the hydrophobic region II.

The potential of phosphate mimetics as kinase inhibitors

AS5 is a relatively flat p110 α /p110 δ dual-selectivity inhibitor with only modest affinities for these two isoforms. Its dimethoxy-aniline group occupies the adenine pocket, where it interacts with the hinge Val828, but does not project deeply into the affinity pocket (Fig. 3d). It is conceivable that modifications on this scaffold that target polar moieties within the affinity pocket could increase the potencies of AS5 derivatives. Coupled to the quinoxaline group is a *p*-fluorobenzenesulfonamide, and when superimposed on the p110 γ -ATP crystal structure it becomes apparent that the sulfonyl group of AS5 roughly co-localizes with the α -phosphate group of ATP. This compound reveals two strategies to mimic the ATP phosphates to achieve inhibition of p110 α and p110 δ . First, one of the sulfonyl oxygens of AS5 is a hydrogen bond acceptor for P-loop Ser754. Second, the fluoro-phenyl group exits the active site close to the DFG Asp911, in proximity of the space occupied by the β - and γ -phosphates in the p110 γ -ATP structure.

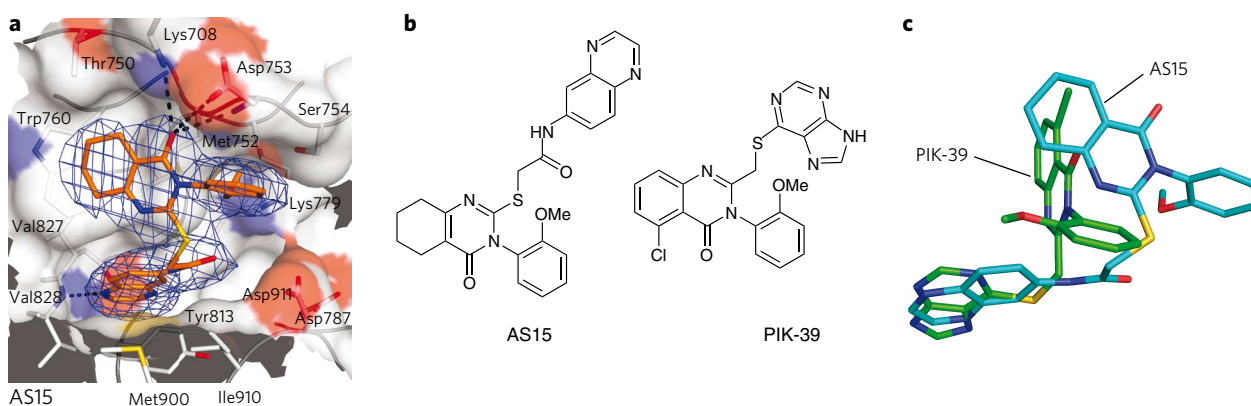


Figure 4 | Binding mode of the p110 δ -selective PI(3)K inhibitor AS15 and comparison of AS15 with the propeller-shaped inhibitor PIK-39. (a) The highly p110 δ -selective compound AS15 does not open the specificity pocket and makes extensive use of a hydrophobic patch between Trp760, Thr750 and Met752 adjacent to the adenine-binding pocket. The $2mF_o - DF_c$ contouring level is 1 σ . (b) Chemical structures of AS15 and PIK-39. (c) Superposition of AS15 and PIK-39 to demonstrate their different modes of binding within the active site of p110 δ .

GDC-0941 uses the space above hydrophobic region II

The identification, characterization and development of the tricyclic pyridofuopyrimidine lead PI-103 (refs. 45–47), a very potent dual-selective PI(3)K/mTOR inhibitor, has led to the pan-selective class I PI(3)K thienopyrimidine inhibitor GDC-0941, which has no off-target activity against mTOR (ref. 32). GDC-0941 is orally bioavailable and currently in phase I trials for the treatment of solid tumors³³.

Its structure in complex with p110 δ (Fig. 3e) confirms the previously described binding mode to p110 γ (ref. 32) but also reveals new features. Whereas the piperazine ring adopts a twisted chair conformation in the p110 γ structure, it is present in a distorted boat conformation in the structure of p110 δ . The terminal methane-sulfonylpiperazine group is also oriented differently in both structures. In p110 δ , this group is marginally tilted with respect to the central thienopyrimidine scaffold and thereby comes closer to the P-loop. Instead of the Lys802-p110 γ (Arg770-p110 α), the Thr750 at the equivalent position in p110 δ is unable to establish a hydrogen bond to the inhibitor's sulfonyl oxygen. However, a different lysine residue (Lys708) interacts with the sulfonyl group of GDC-0941, which indicates why this compound does not lose affinity for p110 δ .

AS15 explores the nonconserved rim of the active site

Although AS15 (13) is chemically related to the quinazolinone purine inhibitor PIK-39, its cocrystal structure with p110 δ reveals an unexpected mode of binding (Fig. 4). Instead of wedging in between Met752 and Trp760, the hexahydroquinazolinone group presses tightly against Met752 (in its 'in' position) and Trp760. By comparing the binding modes of PIK-39 and AS15 to p110 δ , three reasons can be deduced for why PIK-39, but not AS15, is able to induce the specificity pocket. First, whereas the purine group of PIK-39 acts as a hydrogen bond donor and acceptor, the AS15 quinoxaline group interacts only with the backbone amide of hinge Val828. Second, the nonplanar nature of the hexahydroquinazolinone may exceed the capacity of the specificity pocket. In its alternate location, the hexahydroquinazolinone packs into a shallow dimple formed between Met752, the small side chain of Thr750, and Trp760. In other p110 isotypes, the residue equivalent to Thr750 is a lysine or arginine. This interaction may account for the high isotype selectivity of this compound. Third, compared with the shorter thiomethyl linker of PIK-39, the longer methylthioacetamide linker of AS15 might be more conformationally restrained due to the planar nature of the linker's peptide bond. This planarity might prevent the hexahydroquinazolinone from being positioned in a way that would allow for the induction of the specificity pocket.

A number of additional p110 δ -specific interactions are formed in a manner whereby the ketone oxygen from the hexahydroquinazolinone group acts as a hydrogen bond acceptor for the backbone amide of the P-loop Asp753 and for the primary amine of Lys708. The P-loop Asp753 is specific to p110 δ (the corresponding residue is Ser773 in p110 α and Ala805 in p110 γ), and Lys708, which is located outside of the active site, has an equivalent in p110 α (Lys729) but not in p110 γ (Ser 760). Given that AS15 does not occupy the affinity pocket, modifications of the compound exploring this pocket should result in an increased potency for p110 δ .

DISCUSSION

The p110 δ -inhibitor crystal structures presented here show that selectivity can be achieved by exploiting both differences in flexibilities among the isoforms and isotype-specific contacts beyond the first shell of residues that interact with ATP. Flexibility-based inhibitors are generally able to use the inherently greater pliability of the p110 δ P-loop. All propeller-shaped inhibitors create a new specificity pocket not present in the apo form of the enzyme. Small modifications of this framework (as found in AS15) can result in inhibitors that are highly selective by establishing unique p110 δ -specific interactions without the formation of the specificity pocket. The plasticity

of p110 δ may enable this isoform to more readily accommodate even very rigid compounds. Our structures also suggest that introducing moieties interacting with the hydrophobic region II at the mouth of the active site might help to improve pharmacokinetic properties of drug-like PI(3)K inhibitors such as GDC-0941.

Initial molecular dynamic simulations suggest that allosteric pockets such as the specificity pocket can be identified with computational approaches. A similar method that imposes stress on the ATP-binding pocket may identify new strain-prone regions that could be exploited by inhibitors.

The strategy of exploring the affinity pocket is a very powerful approach to augment the potency of inhibitors while maintaining selectivity. Further development of selective inhibitors for other isotypes and for overcoming potential resistance mutations that frequently accompany treatment with inhibitors will require a broader range of PI(3)K and PIKK structures.

METHODS

Construct design, expression and purification of Δ ABDp110 δ . Briefly, the TEV-insertion construct of mouse p110 δ was generated using the overlapping PCR method, digested with BglII and XhoI at sites encoded by the primers and ligated into pFastBac-HTa (Invitrogen) cut with the BamHI and XhoI restriction enzymes (New England Biolabs). The correct insertion of the TEV site was confirmed by DNA sequencing (amino acid sequence: 101-LVARE-(105)-ENLYFQG-(106)-GDRVKK-111). The construct has an N-terminal extension encoded by the vector (MSYHHHHHHHDYDIPTTENLYFQGMADL) preceding the first residue of p110 δ . This extension has a His₆ tag and an additional vector-encoded TEV-cleavage site. Recombinant baculovirus was generated and propagated according to standard protocols. For expression, Sf9 insect cells at a density of 1×10^6 ml⁻¹ were co-infected with an optimized ratio of viruses encoding the catalytic and regulatory subunit. As a regulatory subunit, we used the iSH2 fragment of the human p85 α (residues 431–600), tagged with an N-terminal, noncleavable His₆ tag. The culture was incubated for 48 h after infection, and cells were harvested and washed with ice-cold phosphate-buffered saline (PBS), flash-frozen in liquid N₂ and stored at –20 °C. For purification, cell pellets corresponding to typically 8 l of culture were defrosted and resuspended in 250 ml of buffer A (20 mM Tris pH 8, 100 mM NaCl, 5% (v/v) glycerol and 2 mM β -mercaptoethanol). After addition of 2 tablets of complete EDTA-free proteinase inhibitors (Boehringer), the suspension was sonicated and the lysate was spun at 42,000 r.p.m. for 45 min. The supernatant was filtered through 0.45 μ m filter units (Sartorius) and loaded onto a 5 ml HisTrap column (GE Healthcare). After a wash step with buffer A, the column was eluted using a gradient from 0–100% buffer B (buffer A + 200 mM imidazole). Fractions of the p110 δ -iSH2 complex were pooled and loaded onto a 5-ml heparin column equilibrated with heparin A buffer (20 mM Tris pH 8, 100 mM NaCl, 2 mM β -mercaptoethanol). The column was washed and eluted with a gradient from 0–100% heparin B buffer (heparin A + 1 M NaCl). This chromatography step resulted in a separation of excess His₆-tagged iSH2 (earlier peak) from the p110 δ -iSH2 complex (later peak). The p110 δ -iSH2 fractions were pooled and adjusted to 5 mM β -mercaptoethanol. TEV proteinase at a w/w ratio of 1:10 was added, and the mixture was incubated overnight at 4 °C. After verifying that the cleavage reaction was complete, the solution was adjusted to 30 mM imidazole and passed over a second 5-ml HisTrap column to remove the ABD/His₆-iSH2, and Δ ABDp110 δ was collected in the flow-through. Following a concentration step using Vivaspin 20 concentrators with a 50 kDa MWCO (Vivascience), the protein was subjected to gel filtration on an S200 16/60 HiLoad column (GE Healthcare) and eluted in 20 mM Tris pH 7.2, 50 mM (NH₄)₂SO₄, 1% (v/v) ethylene glycol, 1% (w/v) betaine, 0.02% (w/v) CHAPS and 5 mM DTT. Finally, fractions were pooled and concentrated to 4.5–5 mg ml⁻¹ as determined spectrophotometrically using the extinction coefficient 129,810 M⁻¹ cm⁻¹ at 280 nm, flash frozen in liquid N₂ and stored at –80 °C. We have applied this strategy to all other class IA isoforms (not shown).

Synthesis and characterization of SW13, SW14, SW30, DL06 and DL07.

A detailed description of the synthesis and characterization of these compounds can be found in the **Supplementary Methods**.

X-ray crystallography. High-quality diffraction data of Δ ABDp110 δ crystals grown in the presence of inhibitors were obtained using a microseeding protocol implemented on our robotic setup. All crystal structures were solved by molecular replacement. See **Supplementary Methods** for additional details.

Lipid kinase activity assay. To compare the PI(3)K lipid kinase activity of the crystallized mouse Δ ABDp110 δ construct with that of the full-length mouse p110 δ -mouse p85 α complex and the mouse p110 δ -human p85 α nicSH2 construct, a Transcreeper ADP assay (Bellbrook Labs) was performed according to the manufacturer's instructions. Briefly, for the generation of the ADP/ATP

standard curve, 10 μ l of a 60 μ M ADP/ATP (2 $\times$) mixture of various ADP:ATP concentrations were mixed with 5 μ l of anti-ADP antibody at 80 μ g ml⁻¹ (4 $\times$) and 5 μ l of ADP Alexa633 tracer at 40 nM (4 $\times$) in a low-volume, black and round bottom 384-well plate (Corning). The plate was protected from light and shaken at 500 r.p.m. for 1 h before polarization measurements using a PHERAstar (BMG Labtech) fluorescence polarization microplate reader (λ_{exc} = 612 nm, λ_{em} = 670 nm). For the kinase reaction, 10 nM of enzymes were incubated for 1 h at 25 °C in a buffer consisting of 50 mM HEPES (pH 7.5), 4 mM MgCl₂, 2 mM EGTA and 30 μ M diC₈-PIP₂ (Echelon). The reaction was started by the addition of 30 μ M ATP (Sigma-Aldrich, neutralized). The control included the same components with the exception of the diC₈-PIP₂ substrate. The reaction was stopped by mixing 10 μ l of the kinase reaction with 10 μ l of the Stop & Detect buffer (20 mM HEPES pH 7.5, 40 mM EDTA, 0.2% (w/v) Brij-35) containing 20 nM ADP Alexa633 tracer (2 $\times$) and 40 μ g ml⁻¹ ADP antibody (2 $\times$). To allow for signal stabilization, the plate was shaken at 500 r.p.m. for 1 h before fluorescence polarization measurements. The data were plotted and fitted in Kaleidagraph (Synergy Software) using an exponential decay function.

Accession codes. Protein Data Bank: Coordinates of all p110 δ structures and accompanying structure factors have been deposited under the following accession codes: 2WXE (p110 δ -IC87114), 2WXF (p110 δ -PIK-39), 2WVG (p110 δ -SW13), 2WXH (p110 δ -SW14), 2WXI (p110 δ -SW30), 2WXJ (p110 δ -INK654), 2WXX (p110 δ -INK666), 2WXL (p110 δ -ZSTK474), 2WXM (p110 δ -DL06), 2WXN (p110 δ -DL07), 2WYO (p110 δ -AS5), 2WXP (p110 δ -GDC-0941), 2WXQ (p110 δ -AS15) and 2WXR (apo-p110 δ). The structure of the p110 γ -ATP complex (entry 1E8X) was deposited as part of a previous study.

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Author contributions

A.B. expressed and purified the Δ ABDp110 δ construct, crystallized the first Δ ABDp110 δ -inhibitor complexes, collected datasets, determined and refined their structures and performed the kinase activity assay. S.M. helped in the purification, crystallization and structure determination and refinement of several Δ ABDp110 δ -inhibitor complexes. O.W., D.D.L. and B.T.H. synthesized and characterized the inhibitors SW13, SW14, SW30, DL06 and DL07 with input from K.M.S. and determined their IC₅₀ values.

J.I.P. performed the molecular dynamics and free energy perturbation experiments. F.G. devised and provided access to the Morpheus Screen and helped with the implementation of a microseeding protocol. W.-C.H. helped with the insect cell culture and crystal data collection. Y.L. and C.R. designed and characterized the inhibitors INK654 and INK666. P.G., T.R., M.K.S. and J.P.S. synthesized and characterized the inhibitors AS5 and AS15 and helped with large-scale insect cell expression. J.P.S. also provided valuable advice and support throughout the project. R.L.W. helped with the crystal data collection, the structure determination and refinement and the preparation of the movies. The manuscript was written by R.L.W. and A.B.

Competing interests statement

The authors declare competing financial interests: details accompany the full-text HTML version of the paper at <http://www.nature.com/naturechemicalbiology/>.

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